MID-TERM EXAMINATION
(Course Name _B.Tech MAE/ DMAM) (Semester 3rd)
(October, 2023) OFF LINE mode

Subject Code:BMA-205		**Subject: Thermal Engineering I**	
Time : 1 ½Hours			**Maximum Marks : 30**
Note: Q. 1 is compulsory.			
Q1		**(2.5*4)**	
	(a)Explain the Kelvin Planck and Clausius statements. How are they violating each other?		
	(b)Explain the coefficient of performance? What is the relation between COP of refrigerator and COP of heat pump?		
	(c)Define the clausius inequality.		
	(d)Discuss the statement "Entropy is a point function".		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a)The pressure volume correlation for a non flow reversible (quasi static) process is given by p= (8-4V) bar, where V is in m^3. If 150 KJ of work is supplied to the system, determine the final pressure and volume of the system. Take initial volume = 0.6 m^3		
	(b) A perfect gas undergoes the following three separate and distinct processes to execute a cycle. i) constant volume process during which 80KJ of heat is supplied to the gas. ii)constant pressure process during which 85KJ of heat is lost to the surroundings and 20KJ of work is done on it. iii) adiabatic process which restore the gas back to its initial state. Evaluate work done during adiabatic process and the value of internal energy at all the state points if initially its value is 95KJ.		
	(c)A monoatomic ideal gas (y=1.67) and molecular weight = 40 is compressed adiabatically from 0.1MPa, 300K to 0.2 MPa. The universal gas constant is 0.8314 KJ/KMolK. Calculate the workdone during compression of gas in KJ/Kg.		
Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**	
	(a)A heat engine working on a carnot cycle absorbs heat from three thermal reservoir at 1000K, 800K and 600K. the engine does 10KW and rejects 400KJ/min of heat to heat sink at 300K. If the heat supplied by the reservoir at 1000K is 60% of the haet supplied by the reservoir at 600K. Make calculations for the quantity of heat absorbed by each reservoir.		
	(b)Heat flows from a hot reservoir at 900K to another reservoir at 350K. If the entropy change of the overall process is 6.25 KJ/K, Make calculation for the heat flowing out of the high temperature reservoir.		
	(c)Air at 1 bar pressure,290K temperature flows steadily at the rate of 120m^3 /hr into a compressor where its pressure and temperature are respectively raised to 15bar and 390K. During compression process, the heat transfer from the compressor is 10 percent of work transfer from the machine. Neglect KE and PE, evaluate the work and heat interactions. Assume that air is a perfect gas.		